

Cardiovascular Risk

Lifestyle, blood pressure and global hypertension context | Final Exam Data Science Project

Problem & significance

Cardiovascular risk is influenced by measurable clinical and lifestyle factors.

The project translates a real-world health problem into a mathematical binary classification task.

Goal: explain risk factors, compare models, and connect patient data with global hypertension context.

Two independent data sources

Kaggle Cardiovascular Disease Dataset: patient-level features and target.

WHO / Our World in Data hypertension prevalence: country-level public-health context.

The two sources demonstrate consolidation across different data granularities.

Hypotheses

H1: Higher BMI and blood pressure increase cardiovascular disease risk.

H2: Physical activity is associated with lower risk.

H3: Multivariable models outperform a single-factor hypertension baseline.

H4: Patient-level blood-pressure signals align with global hypertension patterns.

Methods

Data cleaning with clinical plausibility filters.

Feature engineering: age_years, BMI, pulse pressure, hypertension flag, obesity flag.

EDA: distributions, group risk rates, correlation heatmap.

Models: Logistic Regression and Random Forest.

Mathematical core

Pearson correlation for linear association between health variables and disease risk.

Logistic regression estimates $P(\text{cardio} = 1 \mid \text{features})$.

ROC-AUC evaluates ranking quality across probability thresholds.

Random Forest captures non-linear relationships and interactions.

Evaluation plan

Train/test split with stratification.

Metrics: accuracy, precision, recall, F1-score, ROC-AUC.

5-fold cross-validation for stability.

Confusion matrix and ROC curves for model interpretation.

Expected insights

Blood pressure and age should be among the strongest risk indicators.

BMI, cholesterol and glucose should add predictive signal.

Physical activity should show a protective relationship.

Global hypertension data supports the public-health relevance of the analysis.

Limitations

Observational data cannot prove causality.

Some medical variables are categorical rather than exact lab values.

Patient-level and country-level datasets cannot be merged row-by-row.

External validity depends on dataset representativeness.

Conclusion

The project demonstrates a complete data-science workflow.

It combines analysis, math, code quality, model evaluation and external context.

The final repository is structured as a portfolio-ready GitHub project.